BANOTH ABHINAV | 26IM61R06

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Percentage
2026–28	M.Tech, Operations Research and Data Analytics	IIT Kharagpur	Pursuing
2025	B.Tech, Mechanical Engineering	National Institute of Technology Calicut	6.53 / 10
2021	Class XII, TSBIE (MPC)	Sri Chaitanya Junior College, Hyderabad	92.8%
2019	Class X, CBSE	St. Peter's Central Public School	74.0%

PROJECTS

Robust Airport Gate Assignment under Delay Uncertainty | GitHub *Python, Gurobi, scikit-learn*

- Built a **predict, simulate, optimise** pipeline on synthetic hub data (Poisson arrivals, gamma delays): **gradient boosting** forecasts arrival delays from 120 past days (MAE 13.97 min, cross-validation grouped by day)
- Turned the forecast into **1,000 delay scenarios** by residual bootstrap and solved a **stochastic MILP** (Şeker & Noyan, 2012) in **Gurobi**; clique constraints and an **exact scenario aggregation** cut the model from 76,970 to 922 constraints and the solve time from ~48 s to ~3.5 s
- On **300 out-of-sample days**, cut gate conflicts by **81%** vs. the timetable plan (from 0.66 to 0.12 per day; conflict-free days from 53% to 88%); planning on the point forecast alone was 5.5× worse

Bearing Fault Diagnosis with ML and a Local LLM | GitHub *Python, scikit-learn, XGBoost, Ollama*

- Built an **envelope-analysis** pipeline on the MFPT bearing vibration data: 16 time-domain, **envelope-spectrum** (BPFO/BPFI) and band-energy features per 0.25 s window, with the train/test split made by recording
- **SVM** reached **100% accuracy** (macro F1 1.000) on 96 held-out windows, including unseen 250 and 300 lbs loads; XGBoost 99.0%, Random Forest 96.9%
- Added rule-based severity (kurtosis, envelope ratios) and a local **llama3.2 LLM** (Ollama) that writes maintenance reports; computing every number and decision in Python removed the LLM's race mix-ups, wrong numbers and unsuitable advice

Gate Assignment IP – Chong & Nordin (2015) Replication | **In progress** *IBM CPLEX (OPL, DOcplex)*

- Formulated the integer program in **OPL and Python DOcplex**; reproduced all **3 published optimal solutions** exactly after identifying and correcting **2 data errors** in the paper
- Proved **total unimodularity** of the constraint matrix and mapped **8 distinct optimal assignments** across the full range of objective weight ratios

SKILLS AND EXPERTISE

- **Optimization:** Gurobi (gurobipy), IBM ILOG CPLEX (OPL, DOcplex), AMPL; LP/MILP, duality, stochastic programming
- **Programming & Data:** Python (NumPy, Pandas, scikit-learn, Matplotlib, Seaborn), R, Django, Git/GitHub
- **Machine Learning:** Gradient boosting, quantile regression, random forest, isolation forest, SVM, XGBoost, k-means, logistic regression; signal processing (envelope analysis); local LLMs (Ollama)

COURSEWORK INFORMATION

- Operations Research
- Statistical Decision Modelling
- Recommender Systems in E-Business
- Safety Economics
- Applied Multivariate Statistics

POSITIONS OF RESPONSIBILITY

Student Representative | Industrial and Systems Engineering Society, IIT Kharagpur *2026 – Present*

AWARDS AND ACHIEVEMENTS

- Represented **NIT Calicut** at the **Inter-NIT Cricket Tournament three times**
- Runner-up (2nd place), **Kerala Cricket Premier League** (North Zone); played at Calicut district and North Zone Kerala levels

EXTRA CURRICULAR ACTIVITIES

- Badminton at a competitive level; trekking, open-water swimming, adventure sports, chess; hands-on natural farming
- Designed and deployed a personal portfolio website (abhistrikes.github.io) on GitHub Pages